NLP: Reliable Multilingual Question Answering

Github link: https://github.com/Vincent-bot901/NLP.git

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1 Introduction and Shared Protocol

Cross-lingual extractive question answering (QA) is a two part objective: (i) accurately localizing a contiguous answer substring within an English context passage given a non-English question, and (ii) reliably abstaining when the supplied context does not support an answer. We investigate this central question on the (`coastalcph/tydi_xor_rc`) dataset, (Asai et al., 2021; Clark et al., 2020; Muller et al., 2023), focusing on Arabic (`ar`), Korean (`ko`), and Telugu (`te`) queries paired with English (`en`) Wikipedia contexts. Because these non-Latin scripts share virtually no symbols with English, the model must overcome an acute cross-lingual representation barrier.

Shared Evaluation Protocol. All models are fitted exclusively on the training split ($N = 6{,}335$) and evaluated on the validation split ($N = 1{,}155$), reporting metrics both overall and separately by language with sample counts N. Due to severe positive class imbalance, binary answerability is evaluated via unweighted Macro-F_1 alongside class-specific precision, recall, and F_1:

$$\text{Macro-}F_1 = \frac{F_1^{(\text{ans})} + F_1^{(\text{unans})}}{2} \tag{1}$$

For BIO sequence labelling, we report token-level Answer-Token F_1 (treating `B-ANS` and `I-ANS` as positive class 1 and `O` as negative class 0) and strict unnormalized character-span Exact Match (EM). Unanswerable examples are pooled directly into Answer-Token F_1 with all-`O` gold labels (any predicted `B/I-ANS` token on an unanswerable passage penalizes precision as a false positive), and score 1 on EM if and only if the decoded sequence is all-`O`, yielding the empty span (`-1, ""`). For integrated QA, we report SQuAD-v2 normalized EM and token F_1 plus Answerability Macro-F_1 at the validation-tuned threshold τ^*. Due to Google Colab T4 GPU limits, neural models were evaluated under a fixed deterministic seed (111) rather than multi-seed ensembles.

Split	Lang	N	Ans%	Unans%	Q Len (IQR)	C Len (IQR)
Train	ar	2558	90.0	10.0	11.0 (4.0)	131.0 (102.0)
	ko	2422	97.4	2.6	12.0 (4.0)	125.0 (96.0)
	te	1355	96.7	3.3	11.0 (5.0)	119.0 (91.0)
	All	**6335**	**94.3**	**5.7**	**11.0 (5.0)**	**125.0 (97.0)**
Val	ar	415	87.5	12.5	10.0 (4.0)	129.0 (101.0)
	ko	356	94.7	5.3	12.0 (4.0)	121.5 (102.8)
	te	384	75.8	24.2	12.0 (6.0)	152.0 (94.0)
	All	**1155**	**85.8**	**14.2**	**12.0 (5.0)**	**136.0 (103.0)**

Figure 1: Dataset statistics by split and language using the `xlm-roberta-base` SentencePiece tokenizer.

2 Corpus Diagnostics and Tokenization

2.1 Distributional Skew and Telugu Shift

Figure 1 summarizes the corpus diagnostics across splits. The training split exhibits an extreme positive prior (94.3% answerable, reaching 97.4% in Korean), rendering raw accuracy uninformative (85.8% on validation by trivially predicting `True`). Crucially, cross-split comparison reveals a severe covariate shift in Telugu (`te`): unanswerable examples increase from 3.3% in training to 24.2% in validation, while median Telugu context length shifts from 119.0 to 152.0 subwords. Models lacking cross-lingual semantic grounding can overfit to the training prior and suffer high false-positive rates on Telugu.

2.2 Hygiene, Lengths, and Offset Integrity

Auditing confirmed 0 missing values across all core fields (`question`, `context`, `lang`, `answerable`, `answer_start`, `answer`); the optional `answer_inlang` field is unpopulated in 6,285 training and 1,055 validation rows and omitted from core experiments. We detected 10 duplicate *(question, context)* in training set, all korean questions. 0 duplicates in validation set

Tokenized questions are concise (median 11–12 subwords), whereas contexts have medians of 125–

Lang	Rank / Token	English Translation / Role	Count
ar	1. ? (*Arabic*)	Terminal question mark (U+061F)	1,499
	2. _? (*Arabic*)	Space-prefixed question mark	1,057
	3. _	Isolated SentencePiece space marker	653
	4. _m	Interrogative/prep. prefix (*mā/min*)	624
	5. _fī	Preposition ("in / at")	619
ko	1. ?	Terminal question mark	2,420
	2. neun	Topic marker postposition	1,154
	3. _	Isolated SentencePiece space marker	986
	4. eun	Topic marker postposition	985
	5. ga	Subject nominative case marker	691
te	1. ?	Terminal question mark	1,093
	2. _	Isolated SentencePiece space marker	506
	3. _evaru	Interrogative pronoun ("who")	274
	4. _?	Space-prefixed question mark	260
	5. lō	Locative case suffix ("in / inside")	240

Figure 2: Five most frequent SentencePiece subwords in training questions per language ("_" denotes U+2581)

136 subwords ($Q_3 \approx 180$, tails > 350). This empirically justifies setting max_seq_length $= 384$ and doc_stride $= 128$ for XLM-RoBERTa, accommodating the full Q_3 distribution within the 512-token positional limit without truncating spans. Finally, programmatic verification across all 5,972 training and 991 validation answerable examples (6,963 total) confirmed that context[start : start + len(answer)] == answer with 0 failures (100% integrity).

2.3 Subword Tokenization Mechanics

Figure 2 reports the five most frequent training question subwords per language using xlm-roberta-base. Linguistic properties emerge: (i) SentencePiece encodes word-initial whitespace via the meta-symbol "_" (U+2581), guaranteeing lossless string reconstruction; and ii) $> 99\%$ of questions end in ‘¿, precluding punctuation-based abstention heuristics.

3 Core Methods and Incremental Pipeline

3.1 Rule-Based Baselines

We implement two non-learned baselines: A majority-class baseline predicting the dominant training label (True) for all examples, as well as a Deterministic Lexical Overlap rule predicting $\hat{y} = \mathbb{I}(|S_q \cap S_c| \geq 1)$, where S_q, S_c are sets of non-punctuation SentencePiece subwords in the question and context. Because non-Latin scripts share zero symbols with English outside incidental Arabic numerals (e.g., years) or borrowed Latin acronyms, the overlap rule cannot exploit lexical semantics and defaults to predicting unanswerable on 96% of validation instances.

3.2 Structured BIO Span Prediction

We cast span extraction as token-level sequence labelling over English contexts tokenized via Unicode regex (\w+|[^\w\s]), mapping character offsets to {O, B-ANS, I-ANS}. Automated unit tests across all 6,335 training and 1,155 validation examples verified four boundary conditions with zero assertion failures: Answers at character 0 start with B-ANS (796 train, 142 val); Multi-token answers contain both B-ANS and I-ANS (3,584 train, 554 val); Punctuation-adjacent spans preserve boundaries (2,787 train, 451 val); and Unanswerable examples map strictly to all-O (363 train, 164 val). Round-trip conversion achieved 98.68% exact recovery on training (79 sub-token crowd annotation mismatches) and 99.50% on validation (5 mismatches).

We train a linear SGDClassifier (log-loss, $\alpha = 10^{-4}$, balanced class weights, seed 110) over 133,684 sparse features per context token x_i: lowercased word, prefix/suffix n-grams, casing/digit/hyphen flags, symmetric window features (x_{i-1}, x_{i+1}), and a question-conditioning feature (in_question_overlap) firing when x_i matches any alphanumeric anchor in the question. We compare an Always O baseline against three decoders: Independent Argmax, post-hoc BIO Repair (converting illegal leading I-ANS to B-ANS), and Constrained Beam Search ($\beta = 4$), which masks invalid O to I-ANS transitions during expansion.

3.3 Arabic Language Modelling

We evaluate N-gram language models on Arabic (ar) questions (2,558 train, 415 validation) tokenized via \w+|[^\w\s]. Sequences are padded on the left with <s> as conditioning history and terminated with an explicit target token </s>. Retaining training tokens with frequency ≥ 2 and mapping singletons/OOV words to <UNK> yields a shared vocabulary of $|V| = 1{,}748$. Against a Uniform Baseline ($P = 1/|V|$), our group compared three smoothed models ($P(w_i \mid h_i) = \frac{C(h_i, w_i)+\alpha}{C(h_i)+\alpha|V|}$): Add-1.0 Unigram ($N = 1$), Add-0.1 Bigram ($N = 2$), and Add-0.01 Trigram ($N = 3$).

3.4 Learned Answerability Classifiers

We implement two supervised classifiers receiving both question and context:

1. **Interpretable Logistic Regression:** Combines 2,500 TF-IDF word unigrams/bigrams over concatenated

$[Question; Context]$ text with 5 standardized dense features (`q_len`, `c_len`, `len_ratio`, `anchor_overlap_count`, `anchor_overlap_ratio`), trained with ℓ_2 penalty ($C = 1.0$) and balanced class weights.

2. **Recurrent Neural Classifier (BiLSTM):** Uses a joint word vocabulary ($|V| = 25{,}958$, freq ≥ 2) with learned embeddings ($d = 100$; static monolingual vectors were avoided as unaligned coordinate spaces distort cross-lingual distances). A single-layer bidirectional LSTM ($h = 64$) encodes Q ($t = 32$) and C ($t = 256$), concatenating final states $[q_{\text{rep}}; c_{\text{rep}}] \in \mathbb{R}^{256}$ into a 2-layer ReLU MLP with `Dropout(0.3)`. A 1-layer architecture was chosen to prevent overfitting on 6,335 skewed examples. A documented limitation is that `q_h[-2]` extracts the forward state at the padded tensor boundary rather than dynamic `q_lens`, partially attenuating forward representations over zero-padding, whereas backward state `q_h[-1]` terminates at $t = 0$ on true initial tokens.

3.5 Integrated Transformer QA

We fine-tune a single joint multilingual Transformer (`xlm-roberta-base`) across Arabic, Korean, and Telugu on concatenated inputs $\mathbf{x} = \langle s \rangle\, Q\, \langle /s \rangle \langle /s \rangle\, C\, \langle /s \rangle$. A linear QA head projects final hidden states $T \in \mathbb{R}^{L \times 768}$ to start and end logits (s_i, e_j), supervising unanswerable windows at the $\langle s \rangle$ index (0). Candidate context spans are scored via $S(i,j) = s_i + e_j$ ($1 \leq i \leq j \leq L$, $j - i + 1 \leq 30$). Following Section 1, we define the no-answer score as the null margin:

$$\text{Score}_{\text{no-answer}} = (s_0 + e_0) - \max_{1 \leq i \leq j \leq L} (s_i + e_j) \quad (2)$$

Tuning over 50 validation quantiles to maximize SQuAD-v2 token F_1 yields $\tau^* = -5.1981$, where $\text{Score}_{\text{no-answer}} > \tau^*$ triggers abstention (`""`).

4 Validation Results and Analysis

Arabic LM Configuration	Order (N)	α	Val Perplexity
Baseline: Uniform LM ($1/\lvert V\rvert$)	0	–	1,748.00
Configuration 1: Unigram LM	1	1.00	81.63
Configuration 2: Bigram LM	2	0.10	**46.11**
Configuration 3: Trigram LM	3	0.01	78.62

Figure 3: Week 38 Validation per-token perplexity

4.1 Language Modelling Perplexity & Hypotheses

As shown in Figure 3, the Add-0.1 Bigram model achieves the lowest validation perplexity (46.11), reducing uncertainty by 97.4% over the Uniform baseline (1,748.00). Although Trigrams ($N = 3$) condition on longer histories, 2-word contexts are exceedingly sparse in a 2,558-question corpus, forcing frequent fallback to uniform smoothing and degrading perplexity to 78.62.

4.2 Answerability and Span Extraction Progression

Figures 4 and 5 trace system performance across architectures:

Linear BIO Bottleneck: In Figure 4, `Always O` achieves 97.47% token accuracy and 14.20% EM purely by abstaining on unanswerable items, proving token accuracy inadequate. Constrained Beam Search ($\beta = 4$) guarantees valid BIO transitions and achieves the highest trained token accuracy (95.64%), 0.1345 Answer-Token F_1, and 10.13% EM. Without cross-lingual embeddings, the linear tagger acts as surface NER guided by English capitalization and shared digits.

W37 Decoder	Lang	N	Tok-Acc	Tok-P	Tok-R	Tok-F_1 / EM
Always O	ar	415	97.34%	0.000	0.000	0.0000 / 12.53%
	ko	356	96.87%	0.000	0.000	0.0000 / 5.34%
	te	384	98.09%	0.000	0.000	0.0000 / 24.22%
	All	**1155**	**97.47%**	**0.000**	**0.000**	**0.0000 / 14.20%**
Indep. Argmax	All	1155	95.58%	0.135	0.138	**0.1365 / 10.22%**
BIO Repair	All	1155	95.58%	0.135	0.138	**0.1365** / 10.13%
Beam ($\beta = 4$)	ar	415	95.66%	0.131	0.112	0.1206 / 9.88%
	ko	356	95.19%	0.160	0.126	0.1409 / 9.55%
	te	384	96.00%	0.121	0.176	0.1435 / 10.94%
	All	**1155**	**95.64%**	**0.135**	**0.134**	**0.1345 / 10.13%**

Figure 4: BIO sequence labeller results on validation ($N = 1{,}155$).

Syntactic Abstention Priors: In Figure 5, Logistic Regression reaches 0.7140 Macro-F_1. Inspecting its learned weights reveals that it exploits **interrogative syntax**: top positive weights target factoid WH-words (`ar` *matā* "when" +3.28, *mā* "what" +3.03, *kam* "how many" +2.28; `ko` *mueos-inga* "what is" +2.93, *eonje* "when" +1.98), whereas boolean particles receive extreme negative weights (`ar` *hal* "is/did" −20.90; `ko` *issneunga* −2.37) because boolean questions rarely map to extractive spans. Among tabular features, `len_ratio` (+1.56) and `anchor_overlap_ratio` (+0.29) predict answerability, while raw `anchor_overlap_count`

Model / Stage	Lang	N	Macro-F_1	Ans-P	Ans-R	Ans-F_1	Unans-P	Unans-R	Unans-F_1
W36: Majority Class	ar	415	0.4666	0.8747	1.0000	0.9332	0.0000	0.0000	0.0000
	ko	356	0.4863	0.9466	1.0000	0.9726	0.0000	0.0000	0.0000
	te	384	0.4311	0.7578	1.0000	0.8622	0.0000	0.0000	0.0000
	Overall	**1155**	**0.4618**	**0.8580**	**1.0000**	**0.9236**	**0.0000**	**0.0000**	**0.0000**
W36: Lexical Overlap Rule	ar	415	0.1320	1.0000	0.0193	0.0378	0.1275	1.0000	0.2261
	ko	356	0.0842	0.8667	0.0386	0.0739	0.0499	0.8947	0.0944
	te	384	0.2373	0.7368	0.0481	0.0903	0.2411	0.9462	0.3843
	Overall	**1155**	**0.1558**	**0.8293**	**0.0343**	**0.0659**	**0.1409**	**0.9573**	**0.2457**
W39: Logistic Regression	ar	415	0.8797	0.9829	0.9504	0.9664	0.7188	0.8846	0.7931
	ko	356	0.5866	0.9538	0.9792	0.9663	0.3000	0.1579	0.2069
	te	384	0.5900	0.7974	0.8385	0.8174	0.3974	0.3333	0.3626
	Overall	**1155**	**0.7140**	**0.9163**	**0.9273**	**0.9218**	**0.5263**	**0.4878**	**0.5063**
W39: BiLSTM Classifier	ar	415	**0.9574**	0.9944	0.9835	0.9889	0.8929	0.9615	0.9259
	ko	356	0.6891	0.9600	0.9970	0.9782	0.8333	0.2632	0.4000
	te	384	0.6113	0.7988	0.9416	0.8644	0.5854	0.2581	0.3582
	Overall	**1155**	**0.7692**	**0.9192**	**0.9758**	**0.9466**	**0.7670**	**0.4817**	**0.5918**
W40–41: XLM-RoBERTa QA	ar	415	0.9381	*SQuAD-v2 EM:* 58.55%			*SQuAD-v2 F_1:* 68.41%		
	ko	356	**0.8676**	*SQuAD-v2 EM:* 57.02%			*SQuAD-v2 F_1:* 67.98%		
	te	384	**0.7630**	*SQuAD-v2 EM:* 51.04%			*SQuAD-v2 F_1:* 59.15%		
	Overall	**1155**	**0.8470**	***SQuAD-v2 EM:* 55.58%**			***SQuAD-v2 F_1:* 65.20%**		

Figure 5: Validation answerability and integrated QA progression across all core models ($N = 1{,}155$).

(-0.25) penalizes incidental matches in long passages. The BiLSTM improves overall Macro-F_1 to 0.7692 (0.9574 on Arabic), but struggles on Korean (0.2632) and Telugu (0.2581) unanswerable recall due to unaligned word embeddings.

Joint XLM-RoBERTa: Fine-tuning `xlm-roberta-base` boosts span extraction by $+45.45$ percentage points over the Week 37 labeller to 55.58% SQuAD-v2 EM and 65.20% F_1, while simultaneously achieving 0.8470 Answerability Macro-F_1 - outperforming the standalone BiLSTM by $+0.0778$ and elevating Telugu Macro-F_1 from 0.6113 to 0.7630.

4.3 Validation Error Taxonomies

BiLSTM Errors): Stratified inspection of 11 validation errors (6 FP, 5 FN across `ar`, `ko`, `te`) revealed four failure modes: (1) *Topical/Entity Distractors* ($4/11$ FP, e.g., `ar` Error #1 matching military terms in Hulagu Khan's Syrian campaign for an Angolan Civil War query; `ko` Error #5 matching "Tesla, Inc."); (2) *Attribute/Role Substitution* ($2/11$ FP, e.g., `te` Error #8 matching the year "2011" and a Georgian minister for a query about Russia's Prime Minister); (3) *Cross-Lingual Script Disjointness* ($3/11$ FN, e.g., `ko` Error #7 on Etruscan rule in Rome with zero Latin anchors; `te` Errors #10–11 on census/Newton dates); and (4) *Boolean Particle Bias* ($2/11$ FN, `ar` Errors #3–4 where initial *hal* triggered false abstention on answerable passages about Saint Paul and Solomon).

Transformer Errors: Across $1{,}155$ validation examples, XLM-RoBERTa made 513 Exact Match errors, categorized in Figure 6 alongside 12 manually inspected cases. Over 83% of errors are extraction failures on answerable passages: Entity Mismatch (45.4%), where cross-attention selects a competing entity of the same semantic type (e.g., extracting *Russia* instead of *England* for cholera's discovery in `te` ID 2), and Boundary Slack (38.0%), where the model locates the exact fact but includes extra dates/units or truncates lists (e.g., *1914 to 1918* vs. *1914* in `te` ID 3; *Egyptian* vs. *Egypt* in `ko` ID 286).

W40–41 Category	Count	%	Representative Validation Examples
Entity Mismatch	233	45.4%	`te` ID 2 (Cholera origin: *England* → *Russia*); `ko` ID 284 (Nordic states: *five* → *27 million people*); `ar` ID 649 (Finnish education clause).
Boundary Slack	195	38.0%	`te` ID 3 (WWI start: *1914* → *1914 to 1918*); `ar` ID 641 (Baltic list → *Finland*); `ar` ID 651 (*1948* → *30 January 1948*); `ko` ID 287 (*999.3 km2* → *385.8 sq mi (999.3 km2)*).
Spurious Extract.	45	8.8%	High topical density drops $\text{Score}_{\text{no-answer}} < -5.20$ on unanswerable context.
Over-Abstention	40	7.8%	Diffuse attention on multi-clause queries pushes $\text{Score}_{\text{no-answer}} > -5.20$.

Figure 6: Taxonomy of all 513 validation errors (44.42% error rate) for the core XLM-RoBERTa QA model.

5 Individual Extensions

5.1 Translate-Test QA Pipeline (Vincent Varga)

Falsifiable Hypothesis. Translating non-English questions ($Q_{\text{ar}}, Q_{\text{ko}}, Q_{\text{te}}$) into English prior to inference will resolve cross-lingual representation gaps, improving Answerable-Only span extraction on Telugu.

Design and Results. Questions were translated into English via `facebook/nllb-200-distilled-600M` and evaluated through the frozen XLM-RoBERTa model at $\tau^* = -5.1981$. Figure 7 confirms the hypothesis: On answerable Telugu queries, Translate-Test improves Ans-Only EM from 48.80% to 51.20% (+2.40%) and Ans-Only F_1 from 59.50% to 61.86% (+2.36%), resolving boundary slack (e.g., `te` ID 10 trimming *26 January 1926* to *1926*; `te` ID 18 trimming a full clause to *Muhammad*). Across all $1{,}155$ validation items, however, Translate-Test fixed 56 core errors but introduced 77 new errors (e.g., `te` ID 14 shifting Egypt's domain *.eg* to *Latin alphabet*), reducing overall EM by 1.81% ($55.58\% \rightarrow 53.77\%$) and

Answerability Macro-F_1 from 0.8470 to 0.8204.

System / Extension	Lang	N	SQuAD EM / F_1	Ans EM / F_1	Macro-F_1
Core: Direct XLM-R	ar	415	58.55 / 68.41	53.17 / 64.44	0.9381
	ko	356	57.02 / 67.98	55.79 / 67.36	0.8676
	te	384	51.04 / 59.15	48.80 / 59.50	0.7630
	All	**1155**	**55.58 / 65.20**	**52.77 / 63.98**	**0.8470**
Ext 1: Translate-Test	ar	415	58.31 / 67.71	52.89 / 63.63	0.9154
	ko	356	52.81 / 63.36	50.74 / 61.89	0.7868
	te	384	49.74 / 57.81	**51.20 / 61.86**	0.7465
	All	**1155**	**53.77 / 63.08**	**51.66 / 62.52**	**0.8204**
Ext 2: Zero-Shot (ru)	ru	396	**49.75 / 57.79**	–	**0.7849**

Figure 7: Extensions compared against the frozen core XLM-RoBERTa QA model ($\tau^* = -5.1981$).

5.2 Zero-Shot Transfer to Russian (Bertan Turan)

Falsifiable Hypothesis. Transferring the frozen XLM-RoBERTa QA model zero-shot to Russian will incur a moderate 5 to 10 percentage point drop in SQuAD-v2 EM and F_1 compared to seen languages because the linear QA head never adapted to Cyrillic interrogative syntax. However, multilingual pretraining will preserve $> 45\%$ EM (far exceeding the 10.13% supervised linear labeller) and maintain Answerability Macro-$F_1 > 0.75$ under the frozen threshold $\tau^* = -5.1981$.

Design and Results. We evaluated all $N = 396$ Russian validation examples in `tydi_xor_rc` through the frozen core pipeline under the Section 1 shared protocol. As shown in Figure 7, zero-shot Russian achieves 49.75% SQuAD-v2 EM and 57.79% F_1, representing a modest drop of -5.83% EM and -7.41% F_1 relative to the seen-language average ($55.58\%/65.20\%$), directly within the hypothesized 5–10 point window. Notably, zero-shot Russian EM (49.75%) nearly matches supervised Telugu (51.04%), while the frozen threshold $\tau^* = -5.1981$ yields 0.7849 Answerability Macro-F_1, insinuating strong cross-script portability of the null-logit margin.

Frozen Pipeline System	Ans OK	Unans OK	Pairs	Pair Acc
W36: Lexical Overlap Rule	4 / 10	6 / 10	0 / 10	0.0%
W37: Sequence Labeller (Beam)	1 / 10	2 / 10	0 / 10	0.0%
W39: BiLSTM Classifier	**10 / 10**	0 / 10	0 / 10	0.0%
W40–41: Core XLM-RoBERTa QA	9 / 10	0 / 10	0 / 10	0.0%
W43: Translate-Test (NLLB)	**10 / 10**	0 / 10	0 / 10	0.0%

Figure 8: Paired challenge set metrics

6 Paired Challenge Set Evaluation

Design and Validation. We constructed `challenge.json` containing 10 matched pairs (20 records). Each pair shares an identical English Wikipedia context, language, creator, and URL, where the unanswerable question applies a minimal semantic alteration (e.g., *DreamHack Winter* $\rightarrow$ *Summer*, *Terrorists* $\rightarrow$ *Counter-Terrorists*, *born* $\rightarrow$ *married*, *won* $\rightarrow$ *hosted*, *joined* $\rightarrow$ *left*). All 20 records passed `validate_challenge.py` with zero errors.

Diagnostic Failure Analysis. Figure 8 reveals an interesting dichotomy: learned neural models achieve 90%–100% accuracy on answerable items but $0/10$ on minimally altered unanswerable items (0% paired accuracy). Specifically: *Language & Extraction:* Failures do *not* stem from script or span localization; Core XLM-RoBERTa extracted $9/10$ exact answers across `ar`, `ko`, and `te` (missing only `bt_pair5_ans` due to Boundary Slack: *"15 times"* vs. *"15"*, which Translate-Test fixed to achieve $10/10$). Furthermore, on unanswerable counterparts, XLM-RoBERTa extracted the exact same entity span (*Fnatic*, *David Johnston*, *24 June 1987*, *Argentina*) with negative no-answer margins between -19.91 and -27.29—indistinguishable from answerable margins (-21.79 to -29.59, e.g., -25.94 vs. -26.33 on `pair1`) and far below $\tau^* = -5.1981$. This demonstrates that standard extractive QA heads rely heavily on entity-type salience and subject co-occurrence rather than verifying fine-grained relational predicates.

7 Limitations and Conclusion

While unaligned surface rules (0.1558 Macro-F_1) and linear sequence labellers (10.13% EM) fail across disjoint scripts, joint multilingual Transformers effectively bridge non-Latin queries and English contexts (55.58% EM, 0.8470 Macro-F_1) and transfer zero-shot to unseen scripts (49.75% EM on Russian). Nonetheless, three core limitations persist: (i) machine translation cascades improve low-resource Telugu extraction ($+2.40\%$ EM) but degrade abstention via translation drift; (ii) severe cross-split shift in Telugu depresses performance relative to Arabic; and (iii) as exposed by our paired challenge set, logit-margin abstention fails when adversarial predicate perturbations preserve subject entities, motivating natural language inference (NLI) verification heads in future work.

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Contribution Statement

In accordance with course regulations for two-member groups, project responsibilities were divided as follows:

- **Shared Core Pipeline:** Both authors jointly conducted the Week 36 dataset diagnostics, missing/duplicate audits, offset verification, and rule-based baselines; collaborated on the Weeks 40–41 XLM-RoBERTa training and error taxonomy; and co-designed the Week 44 evaluation protocol. In Week 38 (Arabic language modelling), both members shared the Uniform baseline and Unigram implementation, while Vincent implemented the Bigram configuration and Bertan implemented the Trigram configuration.
- **Vincent Varga:** Primary responsibility for the Week 37 BIO sequence labeller and constrained beam decoder, the sliding-window preprocessing pipeline, 5 matched challenge pairs (`pair1`–`pair5` across `ar`, `ko`, `te`), and **Individual Extension 1** (Translate-Test QA pipeline using NLLB-200).
- **Bertan Turan:** Primary responsibility for the Week 39 Logistic Regression and BiLSTM answerability classifiers, SQuAD-v2 logit post-processing and threshold tuning, 5 matched challenge pairs (`bt_pair1`–`bt_pair5` in `ko`), and **Individual Extension 2** (Zero-Shot Cross-Lingual Transfer to Russian).

A Reproducibility Appendix

A.1 Hardware and Software Environment

All experiments were executed on Google Colaboratory using an NVIDIA Tesla T4 GPU (15,360 MiB VRAM) and an Intel Xeon CPU. Software versions recorded from the execution environment: Python `3.13`, `transformers==5.20.2`, `datasets==4.8.5`, `evaluate==0.4.6`, `scikit-learn==1.6.1`, `pandas==2.2.3`, `numpy==2.1.3`, and `torch` (CUDA enabled).

A.2 Seeds, Data Counts, and Hyperparameters

Random seeds were fixed at 111 across Python `random`, `numpy`, PyTorch (`torch.manual_seed(111)`), and Hugging Face `Trainer`, and `110` for `SGDClassifier`. All core models were trained on the full filtered training split ($N = 6{,}335$) and evaluated on the full validation split ($N = 1{,}155$):

- **Week 37 Linear BIO Labeller:** `SGDClassifier(loss="log_loss", alpha=1e-4, class_weight="balanced", max_iter=40, random_state=110)`, 750,684 training tokens, 133,684 features, beam size $\beta = 4$ (training time: < 15 seconds).
- **Week 38 Arabic N-gram LMs:** Trained on 2,558 Arabic training questions, evaluated on 415 validation questions, $|V| = 1{,}748$ (min_freq $= 2$), $\alpha \in \{1.0, 0.1, 0.01\}$ (training time: < 2 seconds).
- **Week 39 Logistic Regression:** `TfidfVectorizer(max_features=2500, ngram_range=(1,2))` + 5 `StandardScaler` tabular features (2,505 total dimensions), `LogisticRegression(C=1.0, class_weight="balanced", max_iter=1000, random_state=111)` (training time: < 5 seconds).
- **Week 39 BiLSTM Classifier:** Joint vocabulary $|V| = 25{,}958$, $d_{\text{emb}} = 100$, 1-layer BiLSTM $h = 64$, MLP $256 \to 64 \to 1$ with `Dropout(0.3)`, `BCEWithLogitsLoss`, Adam ($lr = 10^{-3}$, weight decay 10^{-4}), batch size 32 (train) / 64 (val), 5 epochs, max lengths $q = 32, c = 256$ (training time: ~ 45 seconds on T4 GPU).

- **Weeks 40–41 XLM-RoBERTa QA:** `xlm-roberta-base`, AdamW ($lr = 3 \times 10^{-5}$, weight decay 0.01), batch size 8, 3 epochs ($2{,}463$ steps), max_seq_length $= 384$, doc_stride $= 128$, n_best_size $= 20$, max_answer_length $= 30$, validation-tuned $\tau^* = -5.1981$ over 50 quantiles $[0.05, 0.95]$ (training time: ~ 18 minutes on T4 GPU).

- **Week 43 Extensions:** Vincent used `facebook/nllb-200-distilled-600M` (batch_size $= 16$, max_length $= 128$) into frozen XLM-RoBERTa; Bertan evaluated $N = 396$ Russian (`ru`) validation examples through frozen XLM-RoBERTa.

Declaration of using generative AI tools (for students)

☐ **I/we have used generative AI as an aid/tool** *(please tick)*

☐ **I/we have <u>NOT</u> used generative AI as an aid/tool** *(please tick)*

If generative AI is permitted in the exam, but you haven't used it in your exam paper, you just need to tick the box stating that you have not used GAI. You don't have to fill in the rest.

List which GAI tools you have used and include the link to the platform (if possible):

Example: [Copilot with enterprise data protection (UCPH license), https://copilot.microsoft.com]

Describe how generative AI has been used in the exam paper:

1) *Purpose (what did you use the tool for?)*
2) *Work phase (when in the process did you use GAI?)*
3) *What did you do with the output? (including any editing of or continued work on the output)*

we used it for latex formating, general structure of the report and assigment, helped writing questions for week 44 in the json format, debugging and spell check and reviews for possible improvement,

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